
Vision and Action

Jitendra Malik

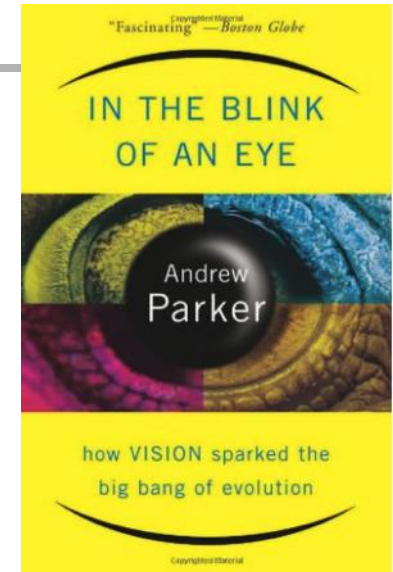
Phylogeny of Intelligence



Cambrian Explosion
540 million years ago

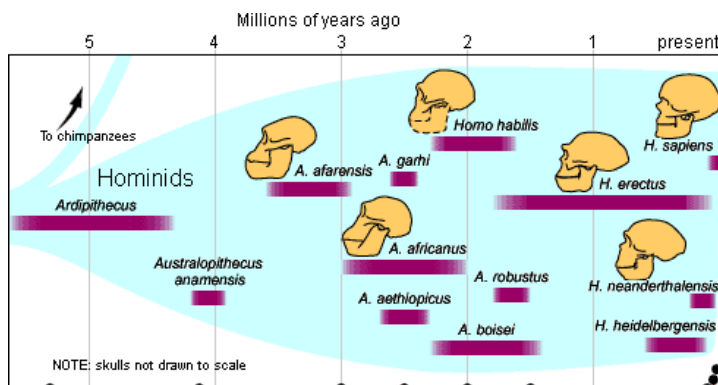
Variety of life forms,
almost all phyla emerge

Animals that could
see and move



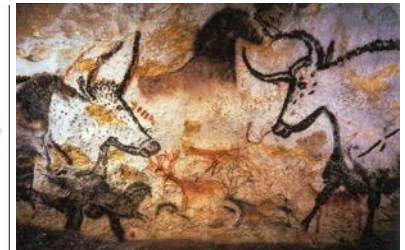
Gibson: we see in order to move and we move in order to see

Hominid evolution, last 5 million years



Bipedalism
Opposable thumb
Tool use

Modern humans, last 50 K years



Language
Abstract thinking
Symbolic behavior

Anaxagoras: It is because of his being armed with
hands that man is the most intelligent animal

Eye Movements while making a cup of tea

Land, Mennie and Rusted (1999)

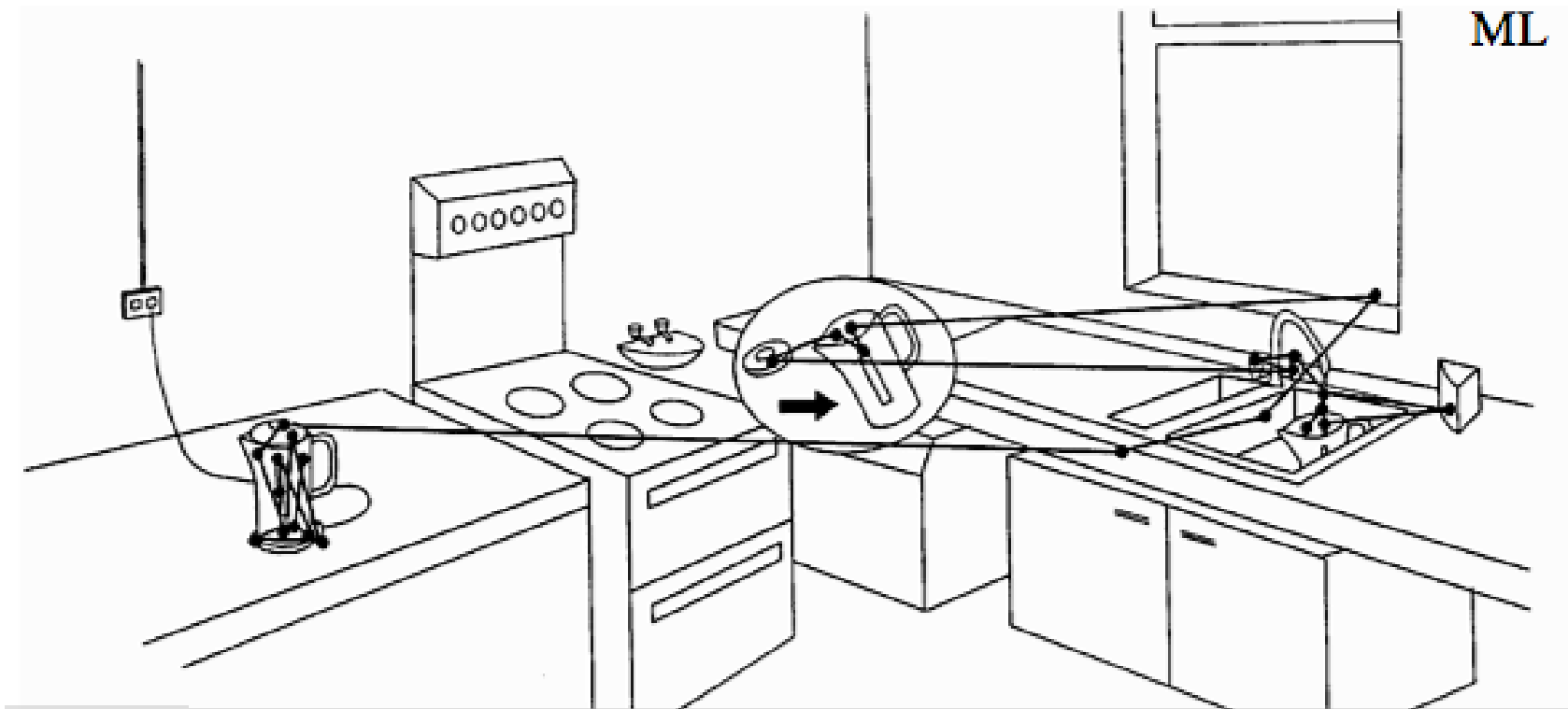
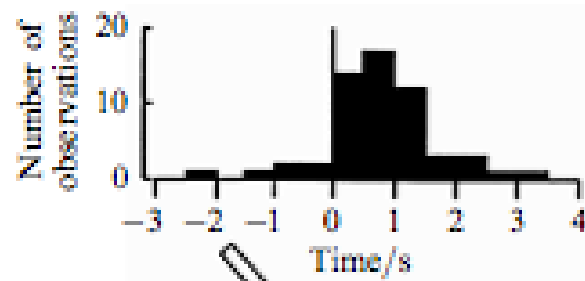




Figure 1. Prints from (a) the activity video, and (b) eye-movement video of the same instant, when the sweetener is dropped into the mug (3.14 on figure 3). The head-mounted camera and

This may be the first example of ego-exo video



Whole body movement (A)

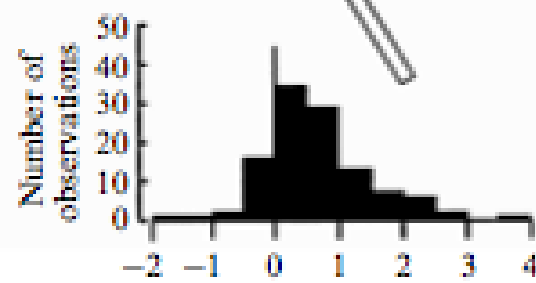
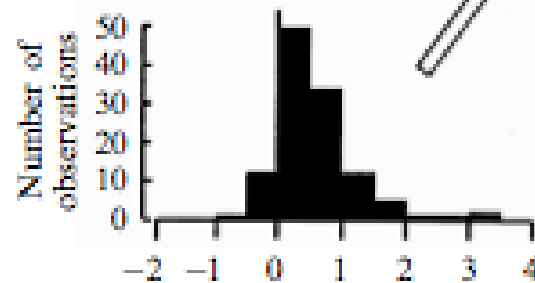
↔ (a) →

Visual fixation of object (B)

↔ (b) →

Manipulation of object (C)

↔ (c) →



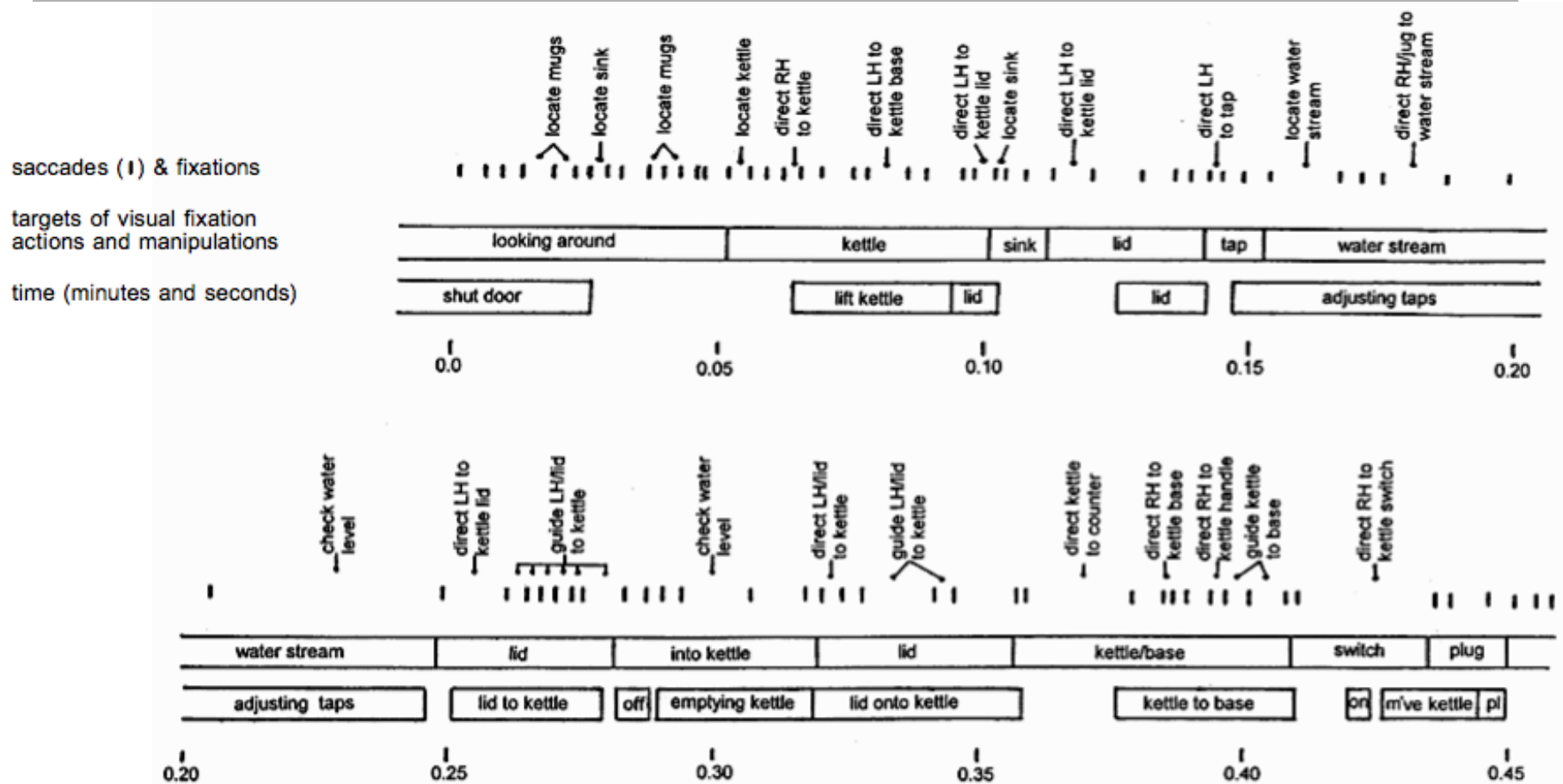


Figure 7. The specific roles of individual fixations in the first level-2 subtask ('fill the kettle') shown in figure 3. The pattern of saccades and intervening fixations are shown, and labels are

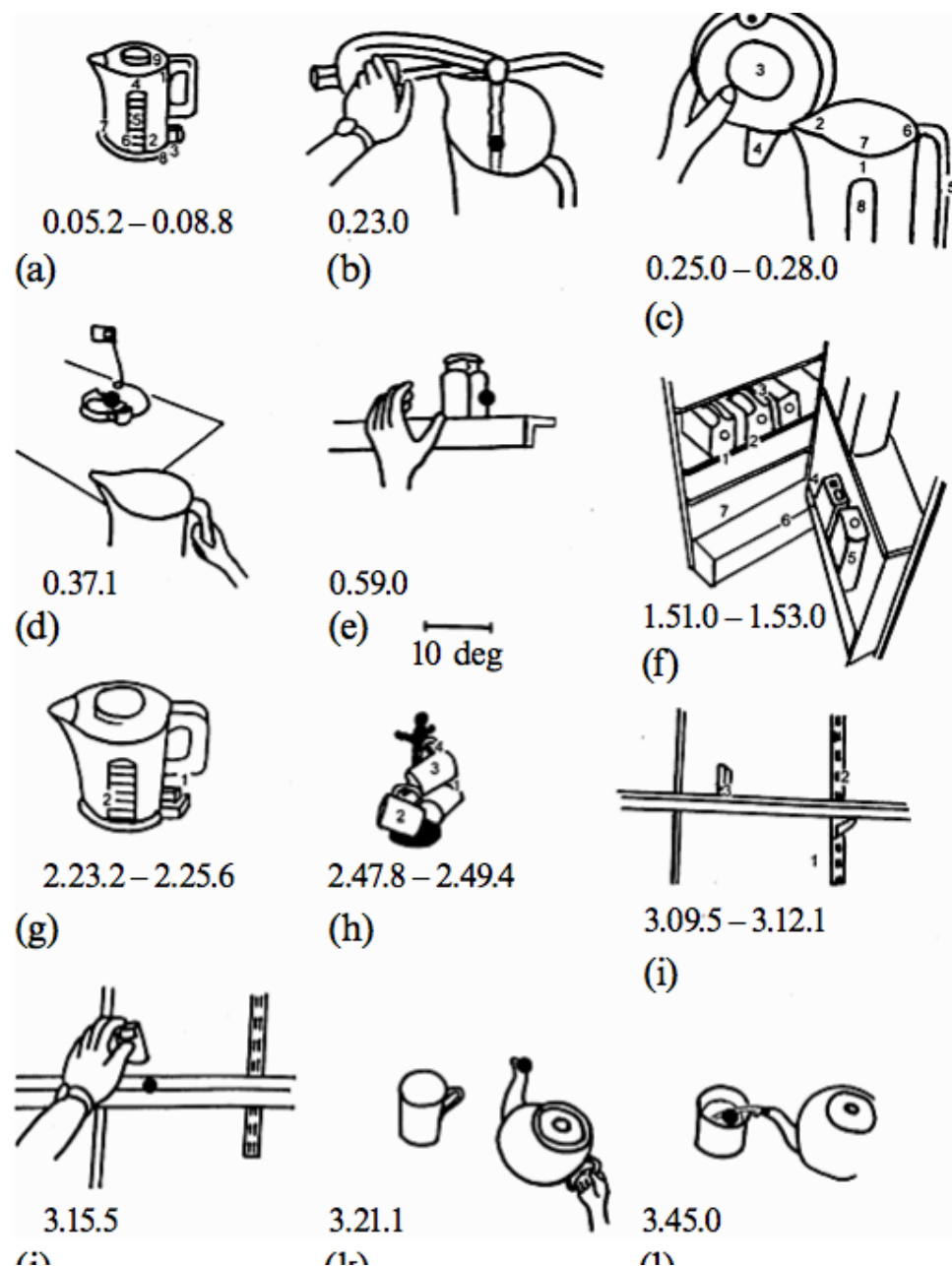


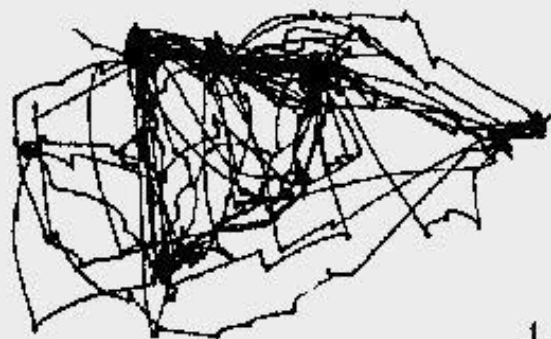
Figure 8. Examples of fixation patterns drawn from the eye-movement videotape. Sequences of successive fixation positions are indicated by numbers on the figures, and single fixations by single black dots. Numbers beneath each figure refer to timings in figure 3. 10 deg scale in centre applies to all figures. (a) Initial examination of kettle. (b) Tap control via water stream. (c) Fitting lid to kettle (drawing made at fixation 4). (d) Moving kettle to base: base is fixated. (e) Hand being directed to the tea-caddy. (f) Search around the inside of fridge 2. The tea-making milk is located at fixation 5. (g) Fixations checking the switch and gauge of the kettle when waiting for it to boil. (h) Selecting a mug. Hand goes to fixation 4. (i) Relocating sweetener prior to use requires 3 fixations. Sweetener last seen 68 s earlier. (j) Replacing sweetener 5 s after (i). Location on shelf is fixated first. (k) Swirling teapot: checking spout. (l) Pouring tea: receiving vessel fixated.

Eyes are directed towards where information that will be useful in the near future is likely to be found

- This has been demonstrated for various everyday tasks such as making tea (Land, Mennie, & Rusted, 1999), making a sandwich (Hayhoe, Shrivastava, Mruczek, & Pelz, 2003), walking (Matthis, Yates, & Hayhoe, 2018) and driving (Land & Lee, 1994; Wilkie & Wann, 2003). It has also been demonstrated for more specialized activities such as reading the score when playing music (Furneaux & Land, 1999).
- In all these cases, gaze precedes and guides movements of the arm or leg. Similar eye movements guide tasks that do not involve such movements. For instance, when reading (Rayner, 1998), searching for an object (Eckstein, 2011), or identifying faces (Peterson & Eckstein, 2013), observers move their eyes to where they anticipate to find the most relevant information at each moment. In general, people anticipate when they will need certain information



Ilya Repin, An Unexpected Visitor, 1884.



Free examination.

1



2

Estimate material circumstances of the family



Give the ages of the people.

3



Surmise what the family had been doing before the arrival of the unexpected visitor.

4



Remember the clothes worn by the people.

5



Remember positions of people and objects in the room.

6



Estimate how long the visitor had been away from the family.

7

3 min. recordings of the same subject

Current Biology

Gaze and the Control of Foot Placement When Walking in Natural Terrain

Highlights

- Gaze and full-body kinematics were recorded during real-world locomotion
- Walkers show distinct gaze strategies appropriate for the demands of each terrain
- Nevertheless, walkers also adopted a constant look-ahead time across all terrains
- Walkers tune gaze behavior to sustain consistent locomotor strategy in all terrains

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In Brief

Matthis et al. recorded subjects' gaze and full-body kinematics during real-world locomotion. Walkers show distinct gaze strategies appropriate for the demands of each terrain. Nevertheless, walkers also adopted a constant look-ahead time across all terrains. Walkers tune gaze behavior to sustain consistent locomotor strategy in all terrains.

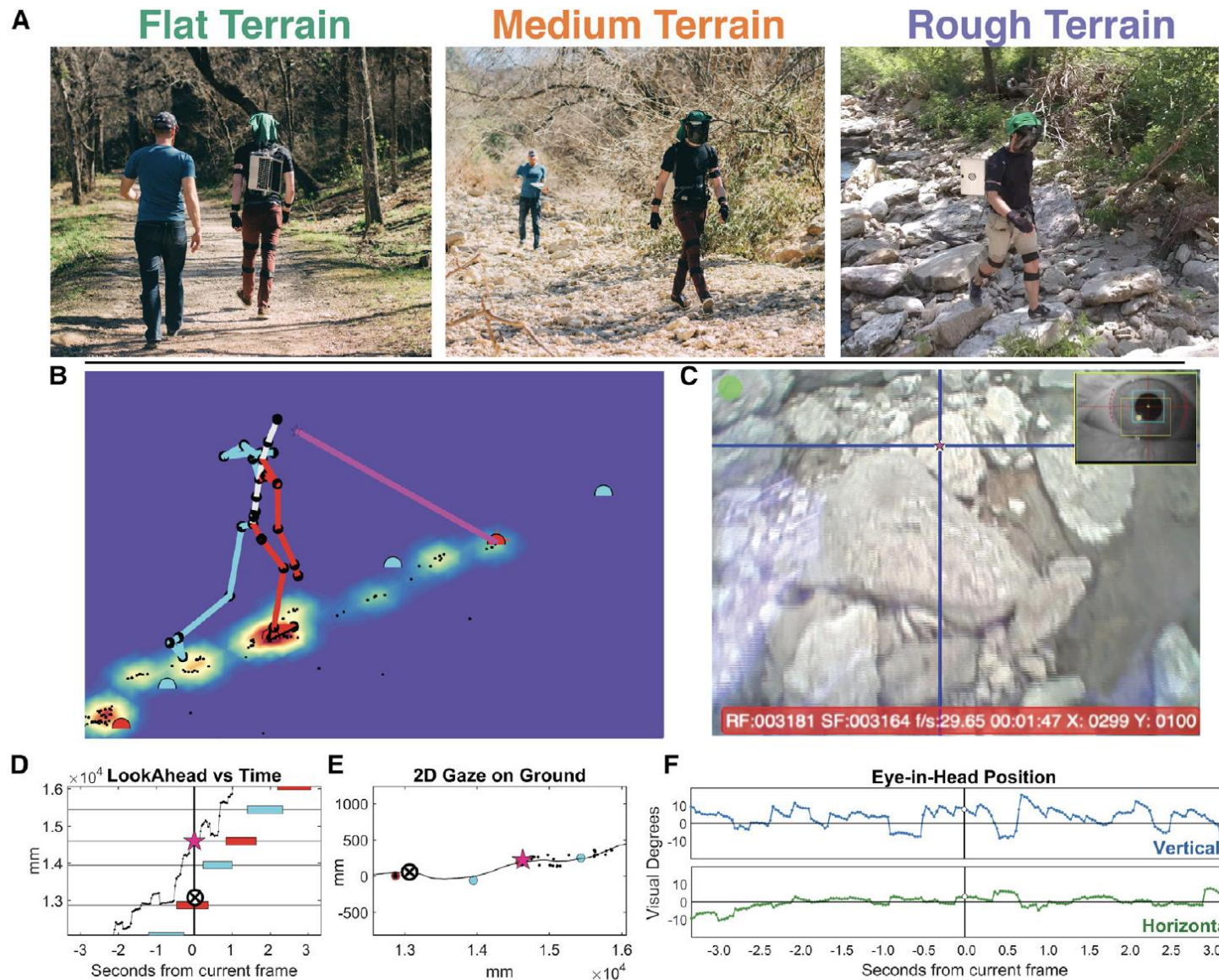


Figure 1. Terrain Types and Data Reconstruction

(A) Subjects walked over three different types of natural terrain: Flat, Medium, and Rough.

Legged Locomotion in Challenging Terrains using Egocentric Vision

CORL 2022



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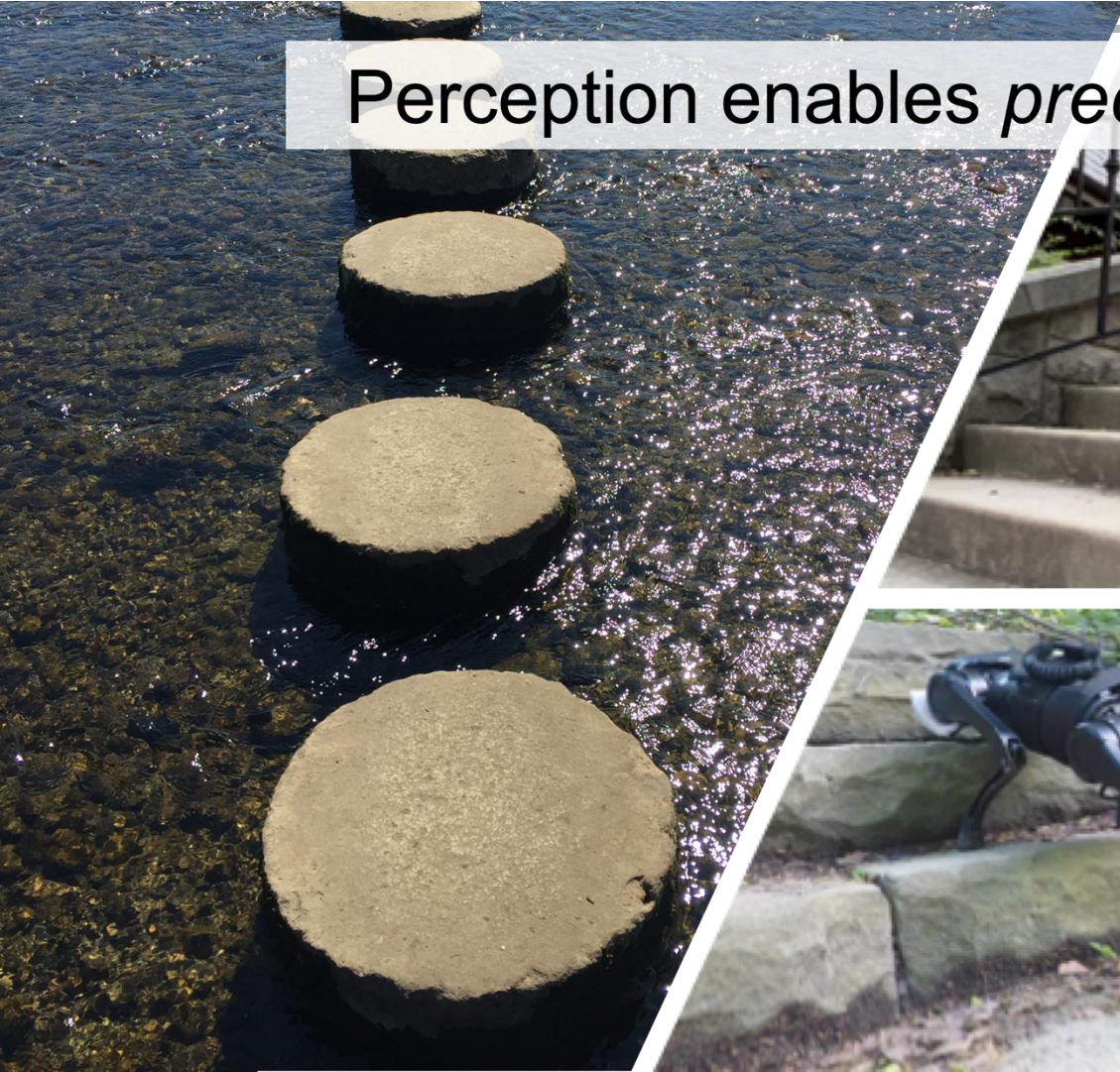
Jitendra Malik†
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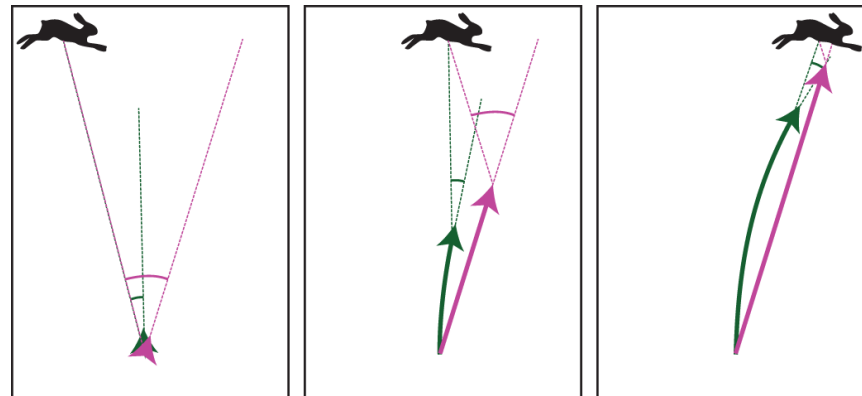
Perception enables *precise* locomotion



Prediction in goal-directed action

Fiehler et al

- **Prediction allows humans and other animals to prepare for future interactions with their environment. This is important in our dynamically changing world that requires fast and accurate reactions to external events.**
- **Knowing when and where an event is likely to occur allows us to plan eye, hand, and body movements that are suitable for the circumstances.**
- **Predicting the sensory consequences of such movements helps to differentiate between self-produced and externally generated movements.**



Keep extent to which one is aiming ahead of target constant by
adjusting movement direction or adjusting movement speed

Figure Legend:

More ways in which interception could be controlled.

How predictive signals drive eye movements when interacting with dynamic objects

- The ability to anticipate future events plays a critical role in many daily contexts. It is particularly evident and important in certain sports. In cricket, for example, skilled batsmen fixate the ball as it leaves the bowler's hand and then quickly make a saccade to near where the ball will bounce, waiting for its image to come back onto the fovea (Land & MacLeod, 2000).
- The brain must have combined the visual information that was available soon after the bowler released the ball with a learned model of the ball's motion to predict the location of the bounce and how the ball will move after the bounce. Saccades sometimes land above a future bounce point (Diaz et al., 2013; Hayhoe et al., 2012), possibly to facilitate tracking the ball after the bounce by directing gaze to where the ball will pass shortly after it bounces.
- Expertise in sports may partly consist of being better at anticipating. Whereas novice baseball batters are usually unable to track the fastest balls with their gaze, elite batters rely on a combination of eye and head rotations to track a fastball (Higuchi, Nagami, Nakata, & Kanosue, 2018). Elite batters can even track the ball up to the moment of bat-ball contact in cricket (Mann et al., 2013).

Efference copy as a form of prediction

- Imagine watching a train pass by. If you do not follow the train with your eyes, its image will sweep across the retina while the image of the context surrounding the train will not. If you do follow the train with your eyes, its image will be (more or less) stable on the retina, while the context's image will produce a motion sweep.
- How does the brain figure out when to attribute retinal motion to object motion and when to attribute it to movements of the eyes?
- Von Holst and Mittelstaedt (1950) proposed that when a motor command is sent to the muscles that move the eyes, a copy of the efferent signal is simultaneously sent to visual areas of the brain.

Predictive signals change perception

- One of the prime examples of predictive sensory attenuation is the phenomenon that we cannot tickle ourselves. Humans experience a touch on their own body as less ticklish if they produce the tactile stimulation themselves (Blakemore, Frith, & Wolpert, 1999).

— AI systems need to build “mental models” —



If the organism carries a ‘small-scale model’ of external reality and of its own possible actions within its head, it is able to try out various alternatives, conclude which is the best of them, react to future situations before they arise, utilize the knowledge of past events in dealing with the present and the future, and in every way to react in a much fuller, safer, and more competent manner to the emergencies which face it (Craik, 1943, Ch. 5, p.61)

Commonsense is not just facts, it is a collection of models

Subthemes

- Predicting sensory changes in the world
- Predicting the consequences of one's own actions
- Combining predictions of eye, arm and object movements.

The Development of Sensitivity of Kinetic, Binocular and Pictorial
Depth Information in Human Infants

Albert Yonas, Carl E. Granrud

Brain Mechanisms of Spatial Vision, pp. 113-145 (1985)

- A program of research is described that explores the development of sensitivity to three classes of spatial information in human infants. The research suggests that sensitivity to kinetic, binocular and pictorial depth information develops in a fixed sequence. Some sensitivity to kinetic information may be present at birth or soon thereafter; sensitivity to binocular information appears between three and five months; and sensitivity to static monocular information appears between five and 7 months of age. These findings may direct research on the development of neural mechanisms that underlie the emergence of responsiveness to spatial information in human infants.